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The Nexus Between Sustainability Uncertainty and Ecological Footprint: Evidence from G7 Countries Using Second-Generation Panel Data Methods

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Abstract

In recent sustainability discussions, the link between policy uncertainty and environmental consequences has received significant attention. This study investigates the impact of sustainability uncertainty on the ecological footprint (EF) in G7 countries between 2002 and 2024. The recently developed ESG-Based Sustainability Uncertainty Index (ESGUI) is used as the primary explanatory variable. The control variables are economic growth, trade openness, and renewable energy consumption. Two complementary estimators capable of addressing cross-sectional dependence, slope heterogeneity, and degrees of mixed integration are applied: CS-ARDL and the AMG estimator. The CS-ARDL results demonstrate that the ESGUI and renewable energy have a weakly significant negative long-term effect on the EF, a result that reverses the positive raw correlation observed in the descriptive statistics, which is driven by the confounding effect of GDP. Moreover, economic growth has a strong positive effect on the EF. Nevertheless, the AMG estimator reveals that the ESGUI is negative but not statistically significant. Furthermore, economic growth contributes to environmental degradation by increasing the EF, while renewable energy and trade openness have a mitigating effect on this EF. Taken together, these results demonstrate that sustainability uncertainty plays only a limited role on average, while economic growth remains the dominant driver of EF. Consequently, policy efforts should focus on decoupling economic growth from resource use, strengthening carbon border regulation mechanisms, and prioritizing fossil fuel substitution rather than simply expanding renewable energy supply.

Keywords: sustainability uncertainty index; ecological footprint; G7 countries; CS-ARDL; AMG



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1. Introduction

Sustainability and sustainable development are among the most debated topics, especially in the last thirty years, in our world, where environmental degradation is increasing. Sustainability is at the very center of the energy, economy, and environment triangle, and is, therefore, discussed across a very wide spectrum, ranging from governments and intergovernmental organizations to politicians, social and natural scientists, and local and international environmental organizations [1]. Although the concept of sustainability is

generally evaluated in its environmental dimension, it should not be forgotten that it has a multi-dimensional (economic, social, and environmental) structure.

Environmental sustainability is the implementation of development models that prioritize environmental policies to reduce environmental degradation and pollution while respecting ecological balance [2]. This understanding points to a similar common goal for every country, regardless of its level of development: to move away from dependence on fossil fuels (FFs) and increase the use of renewable energy (REN) sources to mitigate the consequences of increasing greenhouse gas emissions (GHGs) resulting from the growth of global economic activity. The release of carbon dioxide (CO₂) and other GHGs during the use of FFs such as coal, oil, and gas for energy has led to significant changes in the global climate system [3]. Global CO₂ from the consumption of FFs or various industrial activities has more than doubled in the last half-century, contributing to increased environmental pollution.

While global CO₂ emissions were 16.99 billion tons in 1975, this figure reached 38.09 billion tons by the end of 2023. However, while 8.1 billion tons of the 16.99 billion tons of CO₂ in the same period (1975) originated from the G7 countries, this figure decreased to 7.935 billion tons by the end of 2023 [4]. The most important reason for this can be attributed to countries increasing their investments in REN, which is more environmentally friendly than FF consumption, taking environmental degradation into account. For instance, in 1975, REN consumption as a share of total primary energy consumption was 0.45% in the UK, 1.35% in Germany, 4.74% in the USA, 5.66% in Japan, 8.03% in Italy, and 8.48% in France. Over the past 50 years, these figures have risen to 12.05% in the USA, 12.70% in Japan, 16.47% in France, 20.93% in Italy, 21.29% in the UK, 23.97% in Germany, and finally 27.16% in Canada [4]. Considering that the world average was 14.82% in the same year, many G7 countries are close to or well above the world average. This highlights the importance of investments in REN in addressing CO₂ and other related environmental problems in G7 countries.

The Sustainable Development Goals have been adopted by the United Nations to address climate change and global warming occurring worldwide, and their importance is widely recognized on a global scale [5]. In this context, researchers use various metrics to assess pollution levels and sustainability practices. Two of these metrics are CO₂ and ecological footprint (EF). CO₂ generally measures the volume of GHGs. This metric functions as an indicator of air pollution rather than a direct measurement [6] and deals with the demand side of environmental degradation while highlighting individuals' ability to meet their needs [7]. In contrast, EF analysis consists of six metrics covering forested areas, fishing areas, agricultural lands, carbon footprint, grasslands, and residential areas; therefore, EF tends to provide a broader measure of environmental quality compared with the limited CO₂ scope commonly used in previous studies [8]. EF represents the land area required to sustain the current levels of resource consumption and waste discharge of that population [9]. The continuous increase in the EF poses a serious threat to the future of a sustainable world. This growth leads to deforestation, water scarcity, and biodiversity loss, resulting in numerous environmental disasters. Therefore, well-planned environmental policies by policymakers and coordinated and successfully implemented by governments are essential for achieving better EF outcomes.

There is evidence that planned and implementable rational environmental regulations can improve environmental quality by ensuring sustainability in the G7 group of countries [10]. For example, France signed the Paris Agreement in 2015, committing to controlling the rise in the world's average temperature to +2 °C, and, if possible, +1.5 °C, compared with pre-industrial levels. It has established environmental public policies to achieve a zero-carbon target by 2050. These policies include a maximum annual

average CO₂ reduction of 359 million tons between 2024 and 2028, and improving the overall condition of all groundwater and surface water resources nationwide by 2027 [11]. However, recognizing the need to maintain electricity security and a low carbon footprint in the long term, France is also developing resilient energy system operations while investing in efficiency, six new nuclear power plants, and REN, particularly offshore wind power [12]. Another country, Germany, has already made significant progress in climate action. Under the Climate Action Programme 2030 and the new Climate Action Act, the German government has made a binding commitment to reduce GHGs by 55% by 2030. Setting a target to phase out the use of coal for electricity generation, the German government plans to renovate more buildings to increase energy efficiency [13].

Whether planned environmental policies yield better results or not, in other words, the effectiveness of policies can vary depending on the degree of uncertainty in existing policies. High levels of environmental, social, and governance (ESG)-based uncertainty can produce asymmetric effects on environmental outcomes: while uncertainty may delay environmental investments and undermine policy effectiveness, it can also, under specific conditions, reduce investment in polluting industries and limit energy-intensive consumption [14–16]. Conversely, low policy uncertainty increases the effectiveness of implemented policies. While G7 countries strive to strike a balance in the complex relationship between environmental sustainability and economic growth, diversifying their energy portfolios by adding REN sources such as solar and wind presents both new opportunities and challenges. In this regard, policymakers in authority in G7 countries are called upon to replan their energy subsidies and further increase investments in sustainable energy systems in order to reduce their high EF and create a resilient economic structure in the long term [17].

The transmission channels from sustainability uncertainty to EF operate through regulatory predictability, capital allocation, and energy consumption patterns. When ESG policies become unstable—e.g., due to shifting carbon pricing or inconsistent subsidies—firms face higher risks in committing to long-lived green infrastructure, leading to postponements of low-carbon projects [18]. Such delays slow the replacement of FF-based consumption, keeping the EF higher than under a stable regime [17]. Elevated ESG uncertainty also deters investments in pollution-intensive sectors and reduces total energy demand, partly offsetting the negative impact [19]. Moreover, ESG uncertainty negatively affects green investment in the short run by weakening green signals and investor trust [20]. Consequently, the net effect of the ESG-Based Sustainability Uncertainty Index (ESGUI) on EF is context-dependent, varying with institutional strength and investor responsiveness to policy ambiguity.

This study aims to examine the potential impacts of the sustainable uncertainty index on the EF in G7 countries between 2002 and 2024, as well as to reveal the potential impacts of economic growth, REN, and trade openness (TRADE) on the EF. The motivation for this study, focusing on G7 countries, can be explained by four reasons: First, as expressed in Figure 1, each country included in the G7 group has an EF well above the global average throughout the entire period (1961–2024) for more than 60 years. This high level of environmental pressure, above the global average, makes it appropriate to analyze G7 countries to reveal the factors causing ecological degradation.

Second, as mentioned earlier, G7 economies play a role in the high global CO₂. They have made climate commitments at different levels under various international agreements on environmental sustainability, such as the Paris Climate Agreement and the Kyoto Protocol, aimed at mitigating global climate change and reducing CO₂. Third, these economies are among the leading groups of countries in the global economy in terms of their level of development. Because these countries are deeply integrated into the market, both financially and economically, policy uncertainty arising in one country can easily

spread to others. This makes cross-sectional dependence (CSD) a key concern. Finally, this study reveals the potential impacts of variables on EF using CS-ARDL and AMG estimators, which are second-generation panel methods that take CSD into account. This directly addresses the empirical strategy adopted in this study.

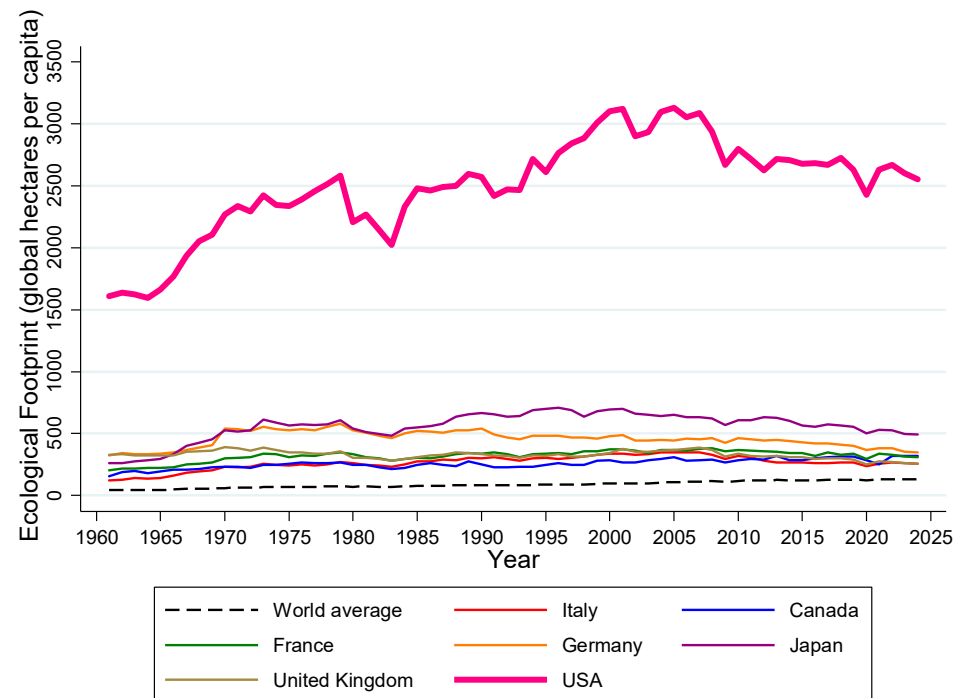


Figure 1. EF of G7 countries and world average over 1961–2024 [21].

This study makes several distinct contributions to the literature. First, to the best of our knowledge, it is the first to examine the impact of the ESGUI on the EF specifically for G7 countries. While previous research has focused on economic policy uncertainty or climate policy uncertainty, the ESGUI captures a broader spectrum of sustainability-related risks. Second, unlike earlier studies that often rely on first-generation panel methods, second-generation estimators (CS-ARDL and AMG) are applied, which are robust to cross-sectional dependence, slope heterogeneity, and mixed integration orders, issues that are particularly relevant for the highly integrated G7 economies.

For the remaining part of this study, Section 2 reviews the specific empirical literature. Section 3 provides information on the data and methodology, and Section 4 evaluates the results obtained from the analysis. In the final stage, Section 5, the main results of this study will be evaluated, the policy recommendations will be discussed, and this study will be concluded with limitations and future possibilities.

2. Literature Review and Theoretical Framework

2.1. Literature Review

The concept of EF, introduced by [22], has become widely accepted as a measure of the pressure of human activities on the biosphere. Unlike one-dimensional factors such as CO₂, EF combines demand across various categories, including arable land, grazing areas, fishing grounds, forest products, carbon sequestration, and settlement areas. This combination provides a more comprehensive picture of environmental degradation [23]. Consequently, an increasing number of studies have attempted to identify the key determinants of EF across different country groups and various time periods. The authors of [24] examined European countries from 1992 to 2020 and found that biocapacity, energy consumption,

industrialization, financial development, life expectancy, and globalization were among the most influential factors. Similarly, for G20 countries, key determinants were identified as economic growth, non-renewable energy consumption, and economic complexity [25].

The relationship between economic growth and environmental degradation has been examined in detail within the framework of the environmental Kuznets curve (EKC) hypothesis. This hypothesis posits an inverted-U-shaped relationship between economic growth and the level of environmental pressure. A meta-analysis by [26] confirmed that the existence of an EKC relationship is sensitive to the characteristics of the data and the choice of environmental indicator. The average turning point was estimated at approximately US\$8910 in annual income. For South Asia, Ref. [27] found evidence supporting both the EKC and the pollution haven hypothesis (PHH) using panel data from 1971 to 2018. Ref. [28] confirmed the inverted-U-shaped pattern between economic growth and environmental degradation in G7 countries.

REN consumption is widely recognized as a key factor in mitigating environmental damage [23]. Researchers examined OECD countries during the period of 1990–2020 and found that REN significantly reduced both EF and CO₂, while economic growth tended to worsen them. Ref. [29] analyzed 27 European countries between 2000 and 2022 and confirmed that REN has a negative and significant effect on EF in both the short and long term. Applying a dynamic panel threshold approach, Ref. [30] reported a negative relationship between REN and EF. However, the magnitude of this effect depends on the share of REN in the total energy mix and the efficiency of renewable technologies.

A recent study by [31] examined the relationship between REN consumption, technological innovations, and EF in G7 countries. This study, using panel cointegration techniques in conjunction with FMOLS and DOLS estimators, found that REN significantly reduces environmental degradation, while population growth and economic expansion increase the EF. The impact of technological innovations was found to differ between countries. Some countries benefit from green technologies, while others experience undesirable increases in resource use due to rebound effects. Ref. [32] applied the MMQR approach, originally proposed by [33], to examine the links between economic growth, REN and non-renewable energy use, energy intensity, FDI, and environmental policy strictness with environmental sustainability (measured by the load capacity factor) in G7 countries. Their results showed that REN consumption, lower energy intensity, and FDI improved environmental sustainability, whereas economic growth and non-renewable energy use degraded it.

The relationship between trade openness and environmental quality is theoretically ambiguous. On the one hand, trade may lead to the relocation of pollutant-intensive industries to countries with weaker environmental regulations; this phenomenon is known as the PHH. Trade may lead to the spread of cleaner technologies and raise income levels, which may increase the demand for environmental quality. For advanced economies, empirical evidence is mixed. Ref. [34] reports that the non-green trade openness index supported PHH, with trade in non-green products associated with lower environmental degradation. Ref. [27] found support for PHH in South Asian economies. However, for high-income countries, some studies show that trade openness tends to reduce the EF; this is likely due to these countries having a comparative advantage in services and cleaner manufacturing [35].

In contrast with the established literature on growth, trade, and energy, the role of policy uncertainty—particularly uncertainty related to environmental and sustainability policies—has only recently begun to receive attention. Economic policy uncertainty has been shown to influence environmental outcomes through various channels. A panel of 12 Organization of Islamic Cooperation countries found that economic policy uncer-

tainty exacerbates environmental degradation by increasing CO₂ [36]. Ref. [37] found that economic policy uncertainty and geopolitical risk weakened the positive impact of economic complexity on environmental quality in G20 countries.

The ESGUI, developed by [38], offers a novel way to measure uncertainty surrounding ESG factors. Ref. [39] uses the ESGUI to examine how sustainability uncertainty affects financial development across 25 countries. A significant negative relationship was found, particularly for financial institutions during periods of low growth. Ref. [40] investigates nonlinear links between the ESGUI and equity markets, oil prices, gold prices, and the VIX index in G7 countries. Their results show that negative shocks have a stronger effect and that equity markets are key determinants of ESG uncertainty.

2.2. Theoretical Framework

The relationship between the ESGUI and clean energy markets can be explained through institutional theory, which emphasizes the role of stable regulatory frameworks and governance structures in shaping investor behavior [41,42]. When ESG policies become unpredictable—due to shifting environmental regulations, inconsistent carbon pricing, or unreliable green subsidies—institutional trust erodes, leading firms and investors to delay or abandon long-term clean energy projects [43]. This institutional fragility amplifies market volatility, particularly in countries with weak governance or frequent policy reversals [44]. Consequently, ESG uncertainty does not merely add noise but actively undermines the credibility of energy transition policies, making clean energy assets more vulnerable during periods of regulatory ambiguity.

The risk–uncertainty framework provides a second lens, distinguishing between measurable risk and Knightian uncertainty [14,16]. ESG uncertainty falls into the latter category: investors cannot assign probabilities to future carbon regulations, technological breakthroughs, or geopolitical shifts in energy policy. This ambiguity increases price volatility and fosters herding behavior, especially in clean energy markets where investments are long-term and policy-sensitive [45]. Unlike conventional risk, which can be hedged, ESG uncertainty creates asymmetric effects—negative impacts dominate during market downturns, while upturns may be less affected—as investors become loss-averse and flee ambiguous assets [46,47].

Finally, resource dependence theory [48] explains how ESG uncertainty disrupts access to external resources such as capital, raw materials, and technology. Clean energy firms rely heavily on government subsidies, supply chains for critical minerals, and international investment flows, all of which become volatile under high ESG uncertainty [49]. This dependence creates fragility, particularly in energy-importing nations, where financing costs rise, and supply security weakens. Consequently, ESG uncertainty can produce two opposing mechanisms: (i) a delay mechanism, where high uncertainty postpones environmental investments and undermines policy effectiveness by eroding institutional trust and increasing financing costs; (ii) a mitigation mechanism, where policy ambiguity reduces investment in polluting industries and limits energy-intensive consumption, potentially lowering environmental pressure under specific conditions. Together, these three theoretical perspectives predict that ESG uncertainty will have heterogeneous, nonlinear, and regime-dependent effects on clean energy markets, varying by market condition (quantile), time horizon, and institutional context [50].

2.3. Research Gap and Contribution

The existing literature has definitively established the importance of economic growth, trade openness, and REN consumption for EF. However, the role of SU—particularly as captured by the recently developed ESGUI—has received limited attention. Further-

more, the theoretical mechanisms outlined above—namely, the delay and mitigation channels—have not yet been empirically tested for the ESGUI–EF nexus in the G7 context. To our knowledge, no previous study has specifically examined the impact of sustainability uncertainty on EF in G7 countries using the ESGUI. Furthermore, most previous studies relied on mean-effects estimators, which can mask significant heterogeneities. This study aims to fill these gaps using CS-ARDL and AMG estimators. Both of these estimators are robust against CSD, slope heterogeneity (SH), and mixed integration degrees. The newly developed ESGUI index adds an additional novelty to the analysis.

3. Data and Methodology

3.1. Data Description and Variable Selection Rationale

There are 161 observations (7 countries $\times$ 23 years) in the dataset with no missing values. Table 1 shows a summary of the variables. The logarithm of GDP and REN are taken to reduce skewness and allow elasticity interpretation. The other three variables are measured as an index (ESGUI), a physical normalized metric (EF in global hectares (gha) per capita), and a ratio (TRADE as a percentage of GDP). Ref. [21] provides EF data for all countries from 1961 to 2022. As of the data access date, the Global Footprint Network reports finalized EF values only up to 2022 due to multi-year reporting lags in underlying UN national accounts. For the years 2023 and 2024, the Global Footprint Network provides nowcast (forecasted) estimates based on extrapolation of recent trends and preliminary indicators. These nowcast values are explicitly flagged by the Global Footprint Network as provisional, and, hence, we include them solely to extend the time coverage to the present.

Table 1. Variable definitions and sources.

Variable	Description	Unit	Source
EF	Ecological footprint	gha per capita	[21]
ESGUI	ESG-Based Sustainability Uncertainty Index	Index	[38]
LGDP	Logarithm of real GDP per capita	Constant US\$ (natural log)	[51]
TRADE	Trade openness (exports + imports)	% of GDP	[51]
LREN	Logarithm of REN consumption	Thousand tons of oil equivalent (natural log)	[51]

The selection of variables in this study is grounded in the established empirical literature on the EF. Following [22], EF is employed as the dependent variable because it provides a multi-dimensional measure of environmental pressure, surpassing single-indicator approaches such as CO₂ emissions [23]. The calculation of EF follows the National Footprint Accounts methodology. The EF measures human demand on ecosystems by tracking the biologically productive land and sea area required to provide the resources a population consumes and to absorb its waste, including carbon emissions [9,21,22]. This demand is calculated by adding imports to and subtracting exports from national production, thereby converting the consumption of thousands of commodities into a standardized unit: gha.

The recently developed ESGUI is the primary explanatory variable, capturing policy and regulatory ambiguity in ESG domains. The ESGUI is constructed following the methodology of [38]. First, ESG sub-indices are created using natural language processing to analyze the frequency of predefined ESG-related keywords in the Economist Intelligence Unit's monthly country reports. Second, an uncertainty sub-index is derived separately using the same text-mining technique on keywords such as “uncertain,” “uncertainty,” and “uncertainties”. Third, the final monthly ESGUI for each country is obtained by av-

eraging the ESG sub-index and the uncertainty sub-index. The index is then normalized to a 0–100 scale to facilitate interpretation across countries and over time. According to [38], the robustness of the ESGUI is empirically supported by its strong correlation with established measures such as the World Uncertainty Index [52] and the economic policy uncertainty index [53].

Economic growth, the logarithm of real GDP per capita (LGDP), is included as a fundamental determinant, as the EKC hypothesis posits an inverted-U relationship between income and environmental degradation. For G7 countries [28], this pattern was confirmed, while [31,32] further demonstrated that economic expansion consistently increases the EF. Hence, higher income levels are associated with higher environmental degradation.

TRADE, the total exports and imports scaled by GDP, is added due to its theoretically ambiguous but empirically relevant effect on environmental quality. The pollution haven hypothesis and technology diffusion channels are well documented [27,34]. For high-income economies, including G7 countries, evidence suggests that TRADE tends to reduce the EF, likely through comparative advantage in cleaner manufacturing and services [28,35].

Note that FF consumption is the primary driver of carbon emissions and, consequently, the EF. However, the theoretical focus of this study is on the energy transition away from FFs rather than on the level of FF use per se. REN consumption directly captures the progress of this transition. The literature widely recognizes the mitigating role of REN consumption, consistently finding that it reduces the EF across different country groups and time periods [23,29–31]. Therefore, a higher REN is expected to lower the EF by reducing dependence on FFs.

3.2. Model Specification

The baseline model in this study is specified in Equation (1). The empirical model can be written in linear form as in Equation (2). Here, α_0 is the constant term, ϕ_1 to ϕ_4 are the slope coefficients to be estimated, and ε_{it} is the idiosyncratic error term. The error term is assumed to have zero mean and finite variance, but no assumption of strict normality or homoscedasticity is made because the estimation methods used are robust to such violations.

$$EF = f(ESGUI, GDP, TRADE, REN) \quad (1)$$

$$EF_{it} = \alpha_0 + \phi_1 ESGUI_{it} + \phi_2 LGDP_{it} + \phi_3 TRADE_{it} + \phi_4 LREN_{it} + \varepsilon_{it} \quad (2)$$

3.3. Econometric Methods

3.3.1. Preliminary Tests

Before proceeding to the main estimation, first, descriptive statistics are presented. Second, the correlation coefficients and variance inflation factors (VIFs) are presented. The VIF of variable j is computed in Equation (3). Here, R_j^2 is the coefficient of determination obtained by regressing the variable j on the remaining independent variables.

$$VIF = 1 / (1 - R_j^2) \quad (3)$$

G7 countries are economically integrated with each other, which can cause CSD. When CSD exists, the estimates will be biased and inconsistent [54]. The CD test, proposed by [55], is performed to find the existence of CSD. The test is robust to unbalanced panels and does not require specifying a particular alternative hypothesis. The CD statistic is computed as

in Equation (4), where N is the number of cross-sectional units, T is the time dimension, and $\hat{\rho}_{ij}$ is the sample estimate of the pairwise correlation of the residuals.

$$CD = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\rho}_{ij} \right) \quad (4)$$

The assumption that slope coefficients are homogeneous across countries is often unrealistic for heterogeneous panels. To test SH, the delta ($\tilde{\Delta}$) test developed by [56] is conducted. The test statistics are defined as in Equations (5) and (6). Here, $\tilde{S}$ is the modified test statistic, and k is the number of explanatory variables (regressors). The first statistic, $\tilde{\Delta}_{SH}$, is the standard delta test statistic, which asymptotically follows a standard normal distribution under the null hypothesis of slope homogeneity. The second statistic ($\tilde{\Delta}_{ASH}$) is a bias-adjusted version that performs better in small samples. $E(\tilde{z}_{it})$ and $var(\tilde{z}_{it})$ denote the mean and variance of the individualized residual terms $\tilde{z}_{it}$, which are derived from the pairwise residuals of the country-specific regressions. Rejecting the null hypothesis indicates that slope coefficients differ across countries.

$$\tilde{\Delta}_{SH} = \sqrt{N} \left(\frac{N^{-1}\tilde{S} - k}{\sqrt{2k}} \right) \quad (5)$$

$$\tilde{\Delta}_{ASH} = \sqrt{N} \left(\frac{N^{-1}\tilde{S} - E(\tilde{z}_{it})}{\sqrt{var(\tilde{z}_{it})}} \right) \quad (6)$$

If, after checking for CSD, the test fails to reject the null of cross-section independence, first-generation unit root tests such as the Im–Pesaran–Shin (IPS) test would be used. Conversely, if CSD is detected, the second-generation cross-sectionally augmented IPS (CIPS) test proposed by [57] is applied. The CIPS test is based on the CADF regression in Equation (7) for each cross-section, where $\bar{y}_t$ is the cross-sectional average of y_{it} (the value of the dependent variable for country i at time t). The CIPS statistic is the average of the individual CADF t-statistics ($CADF_i$), as shown in Equation (8).

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + \gamma_i \bar{y}_{t-1} + \delta_i \Delta \bar{y}_t + \varepsilon_{it} \quad (7)$$

$$CIPS = \frac{1}{N} \sum_{i=1}^N CADF_i \quad (8)$$

To examine the existence of a long-run equilibrium relationship, the panel cointegration test proposed by [58] is applied. This test is robust to CSD and SH. It is based on the error-correction model in Equation (9), where d_t contains deterministic components (e.g., constant or trend), α_i is the adjustment coefficient, and β_i is the vector of long-run coefficients. $x_{i,t}$ is the vector of explanatory variables for country i at time t .

$$\Delta y_{it} = \delta'_i d_t + \alpha_i (y_{i,t-1} - \beta'_i x_{i,t-1}) + \sum_{j=1}^{p_i} \gamma_{ij} \Delta y_{i,t-j} + \sum_{j=0}^{p_i} \theta_{ij} \Delta x_{i,t-j} + \varepsilon_{it} \quad (9)$$

3.3.2. Panel Estimation Techniques: CS-ARDL and AMG

CS-ARDL is developed by [59] in order to evaluate both short-run and long-run parameters while accounting for CSD and SH. This method is applicable independent of whether the variables are I(0) or I(1). The general CS-ARDL (p_x, p_y) specification is

presented in Equation (10). Here, X_{it} and $\bar{Z}_t$ are the cross-sectional averages, and α_i denotes country-specific fixed effects. This estimator is robust to serial correlation, endogeneity, and weak exogeneity of the lagged dependent variable.

$$EF_{it} = \sum_{l=1}^p \lambda_{il} EF_{i,t-l} + \sum_{l=0}^q \beta'_{il} X_{i,t-l} + \sum_{l=0}^s \delta'_{il} \bar{Z}_{t-l} + \alpha_i + \varepsilon_{it} \quad (10)$$

As a second estimator, the AMG estimator developed by [60,61] is used. The AMG estimator is designed to handle CSD and SH, both of which are present in the panel. Unlike standard mean group estimators, the AMG incorporates common dynamic processes (unobserved common factors) into the model. The estimation proceeds in two steps. First, a pooled ordinary least squares regression of the first-differenced dependent variable on the first-differenced regressors and $T - 1$ year dummies is estimated. This allows us to extract the coefficients on the year dummies, which represent the common dynamic process. Second, the estimated common dynamic process is included in each country-specific regression, and the individual country coefficients are averaged to obtain the panel estimates.

The AMG estimator can be expressed in Equations (11)–(13). ΔEF_{it} is the first difference of the EF, ΔX_{it} is the vector of first-differenced explanatory variables (ESGUI, LGDP, TRADE, and LREN), D_t are year dummies, c_t are their coefficients, ϕ_t is the common dynamic process, and the final step averages the country-specific coefficients β_i across countries to obtain the AMG estimates. μ_{it} is the idiosyncratic error term (country-specific random disturbance) for country i at time t . The AMG is robust to non-stationarity and cointegration and provides consistent estimates even if the variables are I(0) or I(1).

$$\Delta EF_{it} = \beta_i \Delta X_{it} + \sum_{t=2}^T c_t \Delta D_t + \varepsilon_{it} \quad (11)$$

$$\phi_t = \hat{c}_t \quad (12)$$

$$EF_{it} = \alpha_i + \beta_i X_{it} + \gamma_i \phi_t + \mu_{it} \quad (13)$$

4. Empirical Findings

This section presents the empirical results obtained from the estimation procedures described in Section 3. All econometric analyses are performed using Stata 14.2.

4.1. Descriptive Statistics and Correlation Analysis

Table 2 reports the descriptive statistics for all variables over the period 2002–2024. The EF ranges from 3.46 to 10.48 gha per capita, with a mean of 5.95 and a standard deviation of 1.80. The ESGUI varies between 14.20 and 63.64, averaging 30.90. Among the control variables, TRADE exhibits considerable variation across countries and years, as indicated by its standard deviation and range.

Table 2. Descriptive statistics.

Variable	Mean	Std. Dev.	Min.	Max.
EF	5.948	1.802	3.460	10.480
ESGUI	30.897	8.607	14.202	63.645
LGDP	10.605	0.182	10.291	11.103
TRADE	53.576	17.142	20.447	88.786
LREN	8.562	0.897	6.316	10.424

Figure 2 displays the time-series evolution of EF for each G7 country from 2002 to 2024. The United States and Canada consistently exhibit the highest EF levels throughout the entire period, ranging between 7 and 10.5 gha per capita, although both show a modest declining trend after the mid-2000s. Second, all G7 countries demonstrate a general reduction in EF over time, particularly visible after the 2008–2009 global financial crisis and again after 2019. Italy, Japan, the United Kingdom, and Germany achieved relatively low EF levels (between 3.5 and 5.5 gha per capita) by the end of the sample period, reflecting improvements in resource efficiency and energy transitions. Third, country-specific trajectories differ: France and Germany show steady declines, while Canada exhibits more volatility. Overall, despite persistently high levels in North America, the downward trends in most G7 countries suggest progress towards reducing ecological pressure.



Figure 2. EF dynamics in G7 countries [21].

Figure 3 presents the evolution of the ESGUI for each G7 country over the period of 2002–2024. The ESGUI ranges from 0 to 100, with higher values indicating greater policy and regulatory ambiguity in ESG domains. The figure reveals substantial cross-country heterogeneity. The United States and Japan exhibit the highest ESGUI values throughout the sample, with the US peaking at approximately 63.6 in 2002 and Japan reaching 51.6 in the same year, both significantly above the G7 average of 31. In contrast, Germany and France display consistently lower ESGUI levels, mostly ranging between 20 and 40, suggesting relatively more stable sustainability policy environments. The UK shows a distinctive pattern: after a period of moderate uncertainty, the ESGUI rises sharply from 2015 onward, peaking at 59.3 in 2019, which likely reflects Brexit-related regulatory ambiguity. Canada and Italy occupy an intermediate position, with values typically between 25 and 50, though Canada experiences a notable spike (52.1) in 2020. A common feature across all countries is the synchronized increase in the ESGUI during the 2008–2009 global financial crisis and again during the 2019–2021 COVID-19 pandemic, indicating that global shocks amplify sustainability uncertainty regardless of country-specific conditions.

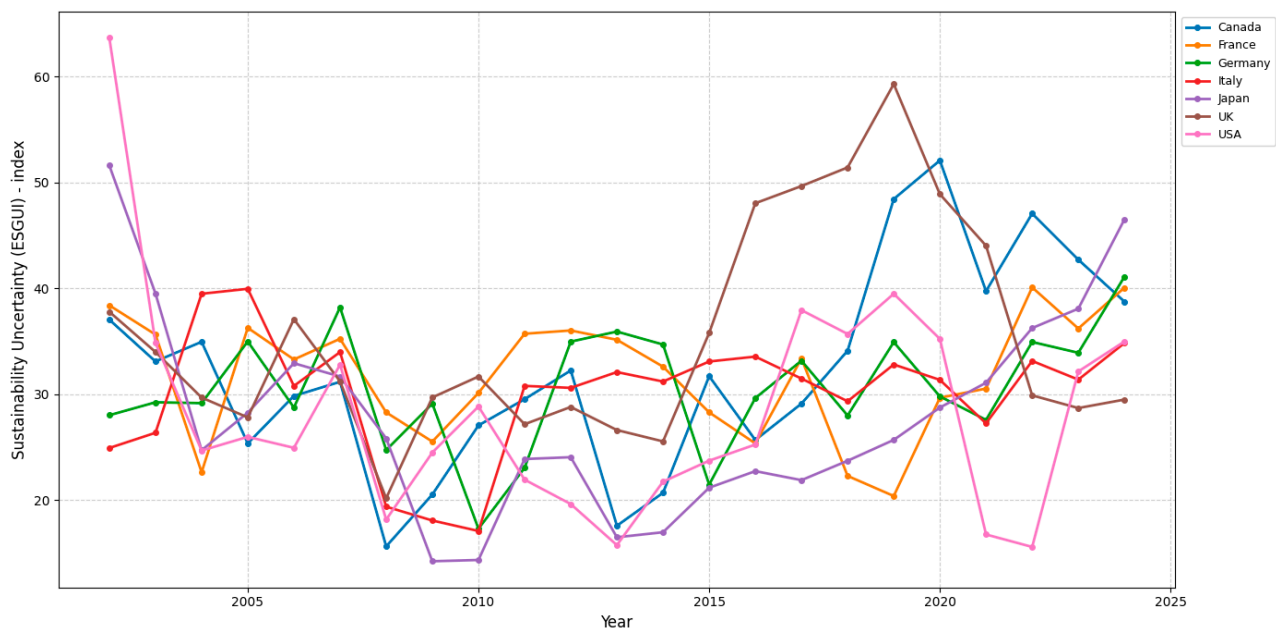


Figure 3. ESGUI dynamics in G7 countries [38].

Table 3 reports the pairwise correlations and VIF values of independent variables. EF is positively correlated with LGDP and LREN and negatively correlated with the ESGUI and TRADE. The correlation coefficients among the independent variables do not exceed 0.57, suggesting that multicollinearity is unlikely to be a problem. This is confirmed by the VIF values reported in Table 3. All VIF values are well below the threshold of 10, indicating no severe multicollinearity. Note that VIFs are computed using a pooled OLS regression as a preliminary diagnostic for severe multicollinearity among the independent variables. This pooled approach is sufficient for detecting structural multicollinearity and does not affect the consistency of the subsequent second-generation panel estimators (CS-ARDL and AMG), which account for cross-sectional dependence and non-stationarity.

Table 3. Correlation matrix and VIFs.

	EF	ESGUI	LGDP	TRADE	LREN	VIF	1/VIF
EF	1.000	−0.088	0.570	−0.256	0.468	—	—
ESGUI	−0.088	1.000	0.062	0.196	0.084	1.05	0.951
LGDP	0.570	0.062	1.000	−0.182	0.274	1.20	0.836
TRADE	−0.256	0.196	−0.182	1.000	0.314	1.27	0.790
LREN	0.468	0.084	0.274	0.314	1.000	1.27	0.788

A noteworthy finding in the correlation matrix is the positive correlation between EF and LREN (0.468), which appears to contradict the standard environmental economics expectation that REN consumption mitigates ecological pressure. This counterintuitive positive correlation flips to a negative coefficient in our multivariate regressions precisely because we control for GDP per capita (LGDP). This direction flip is a classic statistical suppressor effect (a form of Simpson's paradox) due to the confounding role of income. In G7 countries, higher GDP per capita drives two opposing forces simultaneously: on the one hand, it increases overall economic activity, resource consumption, and, thus, the ecological footprint; on the other hand, it enables greater investment capacity for renewable energy infrastructure and policy support for energy transition. Consequently, in the bivariate context, the positive association of both EF and LREN with GDP creates a spurious positive correlation between EF and LREN. Once GDP (and trade openness) are held constant in

the CS-ARDL and AMG regressions, the scale effect is removed, revealing the true partial effect of renewable energy: a net negative (mitigating) impact on the ecological footprint, although only marginally significant. As noted in the literature review [32], some countries experience undesirable increases in resource use due to rebound effects. This positive raw correlation reflects exactly such a rebound effect [62,63]: in G7 countries, REN has often been added to, rather than fully substituted for, expanding FF-based energy demand. In other words, adding renewables to an expanding economy does not automatically lower the ecological footprint unless FFs are actively displaced. Efficiency gains and increased clean energy supply can lower the effective cost of energy services, thereby encouraging higher overall consumption and partially offsetting the environmental benefits. However, the primary driver of the sign reversal is the suppression of the confounding GDP effect, not the rebound effect per se. Once LGDP and TRADE are held constant in the CS-ARDL and AMG regressions, the coefficient of LREN becomes negative (though only marginally significant), confirming that the net long-run effect of REN on EF is indeed mitigating, albeit weak.

4.2. CSD and SH

The results of the CD test by [55] for CSD are reported in Table 4. The null hypothesis of cross-sectional independence is rejected for all variables at the 1% significance level. This confirms the presence of CSD among G7 countries, justifying the use of second-generation panel methods that account for such dependencies.

Table 4. CSD test (Pesaran CD).

Variable	CD Test	<i>p</i> -Value
EF	18.847	0.000
ESGUI	4.827	0.000
LGDP	14.717	0.000
TRADE	9.508	0.000
LREN	10.726	0.000

Table 5 shows the results of the SH test by [56]. Both the delta and the adjusted delta statistics are significant, indicating that the impact of the explanatory variables on the EF varies across G7 countries, further supporting the use of heterogeneous panel estimators.

Table 5. SH test.

Test	Statistic	<i>p</i> -Value
$\tilde{\Delta}_{SH}$	8.161	0.000
$\tilde{\Delta}_{ASH}$	9.492	0.000

4.3. Panel Unit Root Tests

Since there is CSD among G7 countries, the second-generation CIPS unit root test proposed by [57] is applied. Table 6 reports the results for the variables in levels and first differences, including a constant and trend. At levels, the CIPS statistics for EF (−2.174), ESGUI (−2.173), LGDP (−0.768), and TRADE (−2.050) are all greater than the 10% critical value, indicating that these series contain a unit root. For LREN, the CIPS statistic (−2.773) is below the 10% critical value but above the 5% critical value (−2.86), suggesting weak stationarity at the 10% level. After taking first differences, all variables become stationary, with CIPS statistics well below the 1% critical value. Thus, all variables are I(1), except possibly LREN, which is borderline I(0) at the 10% significance level.

Table 6. CIPS panel unit root test.

Variable	Level CIPS	First Difference CIPS
EF	−2.174	−4.778 ***
ESGUI	−2.173	−4.533 ***
LGDP	−0.768	−3.757 ***
TRADE	−2.050	−4.027 ***
LREN	−2.773 *	−5.585 ***

* and *** denote rejection of the null hypothesis at the 10% and 1% significance levels, respectively, based on critical values from [57].

4.4. Cointegration Test

The results of the Westerlund cointegration test by [58] are displayed in Table 7. The test is based on the error correction model and accounts for CSD and SH. The group-mean test G_t is statistically significant at the 5% level (robust p -value = 0.029), providing evidence of cointegration for at least one cross-sectional unit. Overall, the results suggest a long-run equilibrium relationship among the variables.

Table 7. Westerlund cointegration test.

Statistic	Value	Z-Value	p -Value	Robust p -Value
G_t	−2.632	−1.681	0.046	0.029
G_a	−6.855	1.125	0.870	0.173
P_t	−5.993	−1.354	0.088	0.153
P_a	−6.924	−0.293	0.385	0.165

4.5. CS-ARDL Results

Given that the CIPS results reveal a mixture of $I(0)$ (LREN) and $I(1)$ (EF, ESGUI, LGDP, and TRADE) series, together with the presence of CSD and SH, the CS-ARDL estimator of [59] is employed to estimate both short-run and long-run parameters. The results are presented in Table 8. The error correction term is negative and highly significant, confirming the presence of a long-run equilibrium relationship and a rapid speed of adjustment. The CS-ARDL yields two goodness-of-fit measures: the within R^2 (0.58) and the mean group R^2 (0.92). The high mean group R^2 indicates that once cross-country heterogeneity is fully accounted for, the model explains most of the within-country variation in EF, and the substantial difference between the two R^2 values justifies the use of a heterogeneous estimator.

Table 8. CS-ARDL estimation results.

Variable	Coefficient	Std. Error	z	$p > z $
Short run				
L.EF	0.0685	0.0756	0.91	0.364
ESGUI	−0.0070	0.0046	−1.51	0.131
LGDP	5.3886	1.6142	3.34	0.001
TRADE	−0.0043	0.0173	−0.25	0.804
LREN	−0.9647	0.5472	−1.76	0.078
Error correction term	−0.9315	0.0756	−12.33	0.000
Long run				
ESGUI	−0.0072	0.0041	−1.76	0.079
LGDP	6.1663	1.6833	3.66	0.000
TRADE	−0.0038	0.0161	−0.24	0.813
LREN	−0.8796	0.5042	−1.74	0.081
R-squared (within):	0.58	R-squared (mean group):		0.92

In the short-run estimates (not tabulated for brevity), the coefficient of ESGUI is negative but not statistically significant. However, because the error correction term is significant and establishes cointegration, the focus is placed on the long-run coefficients. In the long run, the coefficient of ESGUI is negative and marginally significant, suggesting that greater sustainability uncertainty tends to reduce the EF. LGDP has a positive and significant long-run effect, indicating that economic growth increases environmental pressure. LREN also exhibits a negative and marginally significant coefficient, implying that REN consumption helps to lower the EF. TRADE is not significant in the long run.

To quantify the economic magnitude of ESGUI's influence relative to the control variables, the effects of a one-standard-deviation change in each variable are compared using the long-run coefficients from the CS-ARDL and the standard deviations reported in Table 2. A one-standard-deviation increase in ESGUI (≈ 8.6 units) reduces EF by approximately 0.062 gha per capita. In contrast, a one-standard-deviation increase in LGDP (≈ 0.18 log-points) raises EF by 1.122 gha, while the same increase in LREN (≈ 0.90 log-points) lowers EF by 0.789 gha. Thus, the marginal effect of ESGUI is about one-eighteenth the magnitude of the effect of economic growth, confirming that sustainability uncertainty plays a very limited role on average compared with the dominant influence of income expansion.

4.6. AMG Results

To complement the CS-ARDL findings, the AMG estimator by [60,61] is used. Table 9 reports the AMG results. The AMG coefficient is negative for ESGUI but not statistically significant. LGDP has a positive and highly significant effect. TRADE has a negative and significant coefficient. The LREN coefficient is negative and marginally significant. These results are consistent with long-term CS-ARDL estimations, where ESGUI is marginally significant, and LGDP exhibits a strong positive effect. The AMG findings reveal that the average effect of sustainability uncertainty on the EF is weak or insignificant, while economic growth consistently increases the footprint. The model fit is satisfactory: the root-mean-squared error (RMSE) is 0.1561, which is small relative to the mean of EF (5.95), and the Wald chi2 statistic (54.17; $p = 0.000$) confirms the joint significance of the regressors.

Table 9. AMG estimation results.

Variable	Coefficient	Std. Error	<i>z</i>	$p > z $
ESGUI	−0.0051	0.0036	−1.41	0.158
LGDP	2.6849	0.6861	3.91	0.000
TRADE	−0.0123	0.0052	−2.36	0.018
LREN	−0.6500	0.3542	−1.84	0.066
Constant	−15.2146	4.3617	−3.49	0.000
RMSE (sigma):	0.1561	Wald chi2(4):	54.17 ($p = 0.000$)	

TRADE shows a negative effect that is not significant in CS-ARDL but significant in AMG, indicating a sensitivity to the estimator. The difference in the statistical significance of TRADE between CS-ARDL and AMG can be explained by how each estimator handles unobserved common factors. CS-ARDL models cross-sectional dependence by including lagged cross-sectional averages of the dependent and independent variables, which may absorb some of the long-run variation in trade openness if trade patterns are highly correlated with global cycles. AMG, in contrast, explicitly extracts a common dynamic process (e.g., global demand shocks and oil price fluctuations) and removes it before estimating country-specific coefficients. This allows AMG to isolate the trade-induced technology diffusion effect more clearly, leading to a significant negative coefficient, suggesting that

once common global shocks are filtered out, trade openness exerts a mitigating effect on the ecological footprint in G7 countries.

The observed discrepancy in the statistical significance of ESGUI between the CS-ARDL (marginally significant, $p = 0.079$) and AMG (insignificant, $p = 0.158$) estimators is not unusual in panel econometrics. Past studies that found significant effects of policy uncertainty on environmental degradation often relied on first-generation estimators that fail to account for cross-sectional dependence, slope heterogeneity, and common dynamic processes [19,37]. In contrast, our application of second-generation estimators (CS-ARDL and AMG) explicitly models these complexities, which may explain why our results show a weaker and less robust effect of ESGUI compared with earlier literature.

CS-ARDL accounts for cross-sectional dependence by including cross-sectional averages of the dependent and independent variables as additional regressors, whereas AMG explicitly extracts and includes a common dynamic process. Because the two estimators model unobserved common factors differently, they may produce different levels of precision for coefficients that are economically small and marginally significant. In the current case, the ESGUI coefficient remains negative in both estimators, and the magnitude is similar (-0.0072 in CS-ARDL vs. -0.0051 in AMG). The slight difference in p -values does not reverse the substantive conclusion: the average effect of sustainability uncertainty is weak, economically small, and sensitive to the treatment of common dynamics. This lack of robustness across estimators reveals that the structural stability of sustainability uncertainty across G7 countries is fragile: its effect is not consistently detectable once the common dynamic process is modeled differently. In other words, the relationship between ESGUI and EF is not hard-wired but depends on how unobserved common factors (e.g., global financial cycles, oil price shocks, and pandemics) are accounted for. Therefore, rather than indicating model weakness, the contrast between CS-ARDL and AMG provides valuable insight into the uncertainty surrounding the point estimate and underscores the need for policy recommendations to be based on multiple estimators rather than a single model.

5. Conclusions, Policy Recommendations, and Future Research

The association between environmental degradation and various economic, social, and policy-based factors has long been central to sustainability discussions. Among these factors, uncertainty regarding environmental and sustainability policies has recently attracted attention as a potential determinant of ecological outcomes. While the environmental impacts of economic growth, TRADE, and REN consumption have been extensively studied in the literature, the role of sustainability uncertainty has been relatively limitedly investigated, particularly in the context of developed economies. The G7 countries offer a suitable sample for examining the impact of policy uncertainty on the EFs of these countries.

The present study investigates the influence of sustainable uncertainty, economic growth, renewable energy consumption, and trade openness on the ecological footprint in G7 countries over the period of 2002–2024. The empirical analysis was conducted using a balanced panel dataset (161 observations: 7 countries $\times$ 23 years). The dependent variable was EF, and the primary explanatory variable was ESGUI. Three control variables were also used in the analysis: LGDP, TRADE, and LREN. To account for CSD, SH, and mixed integration degrees, two complementary estimation methods were applied: CS-ARDL and AMG.

5.1. Discussion

The CS-ARDL results confirmed the existence of a long-term equilibrium relationship between the variables with a rapid fit rate. ESGUI exhibited a marginally significant negative long-term effect on environmental pressure; this suggests that higher uncer-

tainty may reduce the EF. This marginally significant negative relationship between ESGUI and EF aligns with the argument that policy uncertainty can, under specific conditions, reduce investment in polluting industries and limit energy-intensive consumption [15]. However, as noted in the introduction, uncertainty may also delay environmental investments and lead to ineffective policy outcomes [14,16], highlighting the asymmetric nature of this relationship.

AMG estimates confirmed the negative but statistically insignificant effect of ESGUI, reinforcing the conclusion that the average effect of sustainability uncertainty is weak. The results of the current study are broadly consistent with [17], who found that the Sustainability Uncertainty Index (SUI) reduces the EF in G7 countries, although the effect was heterogeneous across quantiles. Similarly, using AMG estimators for G7 countries, Ref. [64] found that REN and R&D expenditure reduce environmental pollution, but their results also showed that the EKC hypothesis was not valid for several G7 countries, underscoring the sensitivity of results to country-specific factors.

In contrast, Ref. [19] found that ESG uncertainty significantly increases the EF and reduces the load capacity factor in G7 countries. This divergence may be explained by differences in econometric methodology: Ref. [19] used first-generation estimators that do not fully account for cross-sectional dependence and slope heterogeneity, whereas our study employs CS-ARDL and AMG, which are specifically designed to handle these issues. Moreover, Ref. [37] found that economic policy uncertainty exacerbates environmental degradation in G20 countries, whereas our ESGUI results are weaker, suggesting that ESG-based uncertainty captures a broader set of risks whose link to immediate environmental outcomes may be more indirect and time-lagged.

In terms of control variables, economic growth had a strong positive long-term effect, reflecting the persistent link between income and resource use. This is consistent with the EKC literature for G7 countries [28]. REN consumption showed a marginally significant negative effect, supporting the findings of [23,29] for OECD and European countries. TRADE was not found to be significant in the long run in CS-ARDL; this is likely because the opposing effects cancel each other out in high-income G7 economies, consistent with [35]. However, TRADE showed a negative and significant effect in AMG; this suggests that pollution haven or technology diffusion effects become more pronounced when common dynamic processes are explicitly modeled.

5.2. Policy Recommendations

While our models are not designed for point forecasting, they provide a robust framework for simulating the long-run environmental consequences of policy changes using the estimated coefficients. Given the very weak and only marginally significant average effect of sustainability uncertainty on the EF, no specific policy recommendations are formulated directly targeting ESG-based uncertainty. This statistical finding does not imply that sustainability uncertainty is unimportant; rather, it suggests that despite occasional ESG policy ambiguity, the fundamental determinants of ecological footprint—economic growth, renewable energy, and trade openness—continue to operate as the primary drivers. In other words, while ESGUI serves as an essential control variable demonstrating that sustainability policy shocks do not disrupt the long-run green transition of G7 nations, the heavy lifting of environmental mitigation still rests on decoupling GDP from footprint expansion.

The strong and positive impact of economic growth on EF points to the need for decoupling policies between growth and environmental pressures. In this context, G7 countries should adopt binding resource productivity targets. They should also implement mandatory extended producer responsibility (EPR) schemes and introduce progressive

carbon taxes with revenue recycling for low-income households. Finally, tax burdens should be moved away from labor and onto material footprints and emissions, for example, by taxing resource use rather than labor income.

In addition, the negative impact of TRADE on EF (especially in AMG estimates) suggests that polluting production activities may have shifted outside of G7 countries. This raises the risk of carbon leakage. Therefore, to prevent emissions from shifting geographically and to achieve reductions on a global scale, international cooperation tools such as border carbon adjustment mechanisms need to be strengthened. Meanwhile, the fact that the coefficient of REN consumption, despite being negative, shows only limited statistical significance, reveals that increasing REN use alone is not sufficient. Therefore, the focus of policies should shift from adding REN to the increasing total energy demand to the substitution of FFs. This transformation will be possible through the establishment of binding targets for the phase-out of FFs, the implementation of carbon-pricing mechanisms, and investments in energy storage and grid infrastructure modernization.

5.3. Limitations and Future Research

The study has some limitations. First, the analysis only covers high-income G7 countries; therefore, the validity of the results may be limited for developing or emerging economies. Second, the ESGUI variable is a very broad indicator and does not offer the possibility of separating different sources of uncertainty (e.g., policy, market, or technology-based uncertainties). Third, this study focuses on EF as a single aggregate indicator; its components may respond differently to uncertainty. Fourth, the weak significance of REN consumption necessitates further research using disaggregated data (e.g., solar, wind, and hydro).

There are several extensions of this work as future research. One promising avenue is to examine the role of sustainability uncertainty in other country groups such as BRICS, emerging Asia, or oil-exporting countries. Another extension would be to use alternative environmental indicators such as the load capacity factor (LCF). Additionally, future studies could incorporate interaction terms between sustainability uncertainty and other key variables (e.g., financial development, green innovation, and institutional quality). Finally, the use of more granular data (e.g., at the sectoral or regional level) could help to uncover the mechanisms underlying the weak effect of REN consumption.

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Abbreviations

The following abbreviations are used in this manuscript:

AMG	Augmented Mean Group
CO ₂	Carbon Dioxide
CS-ARDL	Cross-Sectionally Augmented Autoregressive Distributed Lag
CSD	Cross-Sectional Dependence
EF	Ecological Footprint
EKC	Environmental Kuznets Curve
ESG	Environmental, Social, and Governance
ESGUI	ESG-Based Sustainability Uncertainty Index
FFs	Fossil Fuels
GHGs	Greenhouse Gas Emissions
REN	Renewable Energy
SH	Slope Heterogeneity
TRADE	Trade Openness
VIFs	Variance Inflation Factors

References

1. Yeni, O. *Sürdürülebilirlik Ve Sürdürülebilir Kalkınma*. (Ed. Müslüme Narin and Celal Taşdoğan), *Doğal Kaynaklar ve Çevre Ekonomisi*; Gazi Kitabevi: Ankara, Türkiye, 2019; pp. 83–102.
2. Oğuz, İ.H. Politik İstikrar ve Çevresel Sürdürülebilirlik. *Int. J. Bus. Econ. Stud.* **2019**, *1*, 1–11.
3. Deng, S.; Tiwari, S.; Khan, S.; Hossain, M.R.; Chen, R. Investigating the load capacity curve (LCC) hypothesis in leading emitter economies: Role of clean energy and energy security for sustainable development. *Gondwana Res.* **2024**, *128*, 283–297. [CrossRef]
4. Ritchie, H.; Rosado, P.; Roser, M. CO₂ and Greenhouse Gas Emissions. Our World in Data. 2023. Available online: <https://ourworldindata.org/co2-and-greenhouse-gas-emissions> (accessed on 1 January 2026).
5. Caglar, A.E.; Destek, M.A.; Manga, M. Analyzing the load capacity curve hypothesis for the Türkiye: A perspective for the sustainable environment. *J. Clean. Prod.* **2024**, *444*, 141232. [CrossRef]
6. Wang, Q.; Wang, X.; Li, R.; Jiang, X. Reinvestigating the environmental Kuznets curve (EKC) of carbon emissions and ecological footprint in 147 countries: A matter of trade protectionism. *Humanit. Soc. Sci. Commun.* **2024**, *11*, 160. [CrossRef]
7. Huang, Y.; Villanthenkodath, M.A.; Haseeb, M. The nexus between eco-friendly technology and environmental degradation in India: Does the N or inverted N-shape load capacity curve (LCC) hypothesis hold? *Nat. Resour. Forum* **2023**, *47*, 276–297. [CrossRef]
8. Olajuwon, A.O. Does the nexus between remittances, and ecological footprint matter for environmental sustainability in Africa? *J. Acad. Res. Econ. (JARE)* **2024**, *16*, 24–45.
9. Wackernagel, M.; Rees, W. What is an ecological footprint. *Sustain. Urban Dev. Read.* **2004**, *211*, 219.
10. Kartal, M.T.; Mukhtarov, S.; Kirikkaleli, D. Achieving environmental quality through stringent environmental policies: Comparative evidence from G7 countries by multiple environmental indicators. *Geosci. Front.* **2025**, *16*, 101956. [CrossRef]
11. Agence France Trésor. Green OATs: Allocation and Performance Report 2024. 2025. Available online: https://www.aft.gouv.fr/files/medias-aft/3_Dette/3.2_OATMLT/3.2.2_OATVerte/RAPPORT_AFT_OAT%20VERTE_2024_FINAL_WEB_ANG.pdf (accessed on 3 January 2026).
12. IEA. France. 2026. Available online: <https://www.iea.org/countries/france> (accessed on 3 January 2026).
13. The Federal Government of Germany. What Is the German Government Doing for the Climate? 2026. Available online: <https://www.bundesregierung.de/breg-en/service/archive/government-climate-policy-1779414> (accessed on 5 January 2026).
14. Knight, F.H. *Risk, Uncertainty and Profit*; Houghton Mifflin: Boston, MA, USA, 1921; Volume 31. [CrossRef]
15. Pirgaip, B.; Dinçergök, B. Economic policy uncertainty, energy consumption and carbon emissions in G7 countries: Evidence from a panel Granger causality analysis. *Environ. Sci. Pollut. Res.* **2020**, *27*, 30050–30066. [CrossRef]
16. Keynes, J.M. *The General Theory of Employment, Interest, and Money*; Harcourt Brace: San Diego, CA, USA, 1936.
17. Yurtkuran, S.; Atasel, O.Y. Assessing Environmental Sustainability in G7 Countries: The Role of Sustainability Uncertainty Index and Energy R&D Expenditures Using MMQR. *Sustain. Dev.* **2026**, *34*, 848–861. [CrossRef]
18. Alsagr, N.; Syed, Q.R.; Anwar, A. Does ESG Uncertainty Matter for Renewable and Nonrenewable Energy Consumption? Policy Implications in the Context of SDG 7 and SDG 13. *Corp. Soc. Responsib. Environ. Manag.* **2026**, advance online publication. [CrossRef]
19. Qamruzzaman, M. Environmental sustainability in G7: Nexus between digitalization, green innovation, environmental taxes, and ESG uncertainty. *Innov. Green Dev.* **2026**, *5*, 100329. [CrossRef]

20. Kwilinski, A.; Lyulyov, O.; Pimonenko, T. The role of green finance in attaining environmental sustainability within a country's ESG performance. *J. Innov. Knowl.* **2025**, *10*, 100674. [\[CrossRef\]](#)
21. Global Footprint Network. Total Ecological Footprint of Country's Population (in Global Hectares). 2025. Available online: <https://data.footprintnetwork.org/> (accessed on 10 March 2026).
22. Wackernagel, M.; Rees, W. *Our Ecological Footprint: Reducing Human Impact on the Earth*; New Society Publishers: Gabriola, BC, Canada, 1998; Volume 9.
23. Akpanke, T.A.; Deka, A.; Ozdeser, H.; Seraj, M. Ecological footprint in the OECD countries: Do energy efficiency and renewable energy matter? *Environ. Sci. Pollut. Res.* **2024**, *31*, 15289–15301. [\[CrossRef\]](#) [\[PubMed\]](#)
24. Guliyev, H. Determinants of ecological footprint in European countries: Fresh insight from Bayesian model averaging for panel data analysis. *Sci. Total Environ.* **2024**, *912*, 169455. [\[CrossRef\]](#) [\[PubMed\]](#)
25. Raza, A.; Habib, Y.; Hashmi, S.H. Impact of technological innovation and renewable energy on ecological footprint in G20 countries: The moderating role of institutional quality. *Environ. Sci. Pollut. Res.* **2023**, *30*, 95376–95393. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Sarkodie, S.A.; Strezov, V. A review on environmental Kuznets curve hypothesis using bibliometric and meta-analysis. *Sci. Total Environ.* **2019**, *649*, 128–145. [\[CrossRef\]](#) [\[PubMed\]](#)
27. Ozturk, I.; Farooq, S.; Majeed, M.T.; Skare, M. An empirical investigation of financial development and ecological footprint in South Asia: Bridging the EKC and pollution haven hypotheses. *Geosci. Front.* **2024**, *15*, 101588. [\[CrossRef\]](#)
28. Raza, S.A.; Shah, N. Testing environmental Kuznets curve hypothesis in G7 countries: The role of renewable energy consumption and trade. *Environ. Sci. Pollut. Res.* **2018**, *25*, 26965–26977. [\[CrossRef\]](#) [\[PubMed\]](#)
29. Mohsin, M.; Naseem, S.; Baig, S.A.; Rehman, M.Z.U. Gauging the Influence of Renewable and Nonrenewable Energy, Forestry Expansion, and Economic Development on Ecological Footprints: A Study of 27 European Nations Toward Achieving the Sustainable Development Goals. *Int. J. Energy Res.* **2026**, *2026*, 2809292. [\[CrossRef\]](#)
30. Azimi, M.N.; Rahman, M.M. Renewable energy and ecological footprint nexus: Evidence from dynamic panel threshold technique. *Heliyon* **2024**, *10*, e33442. [\[CrossRef\]](#) [\[PubMed\]](#)
31. Türk, M.M. The impact of renewable energy and technological innovations on ecological footprint: Evidence from G-7 countries. *Environ. Res. Technol.* **2026**, *9*, 64–79. [\[CrossRef\]](#)
32. Aybudak, H.G.; Khalid, W.; Usman, M.; Seraj, M.; Rafay, A.; Ali, Q.S. Assessing the impacts of economic growth, stringent environmental policies, renewable energy, and non-renewable energy on environmental sustainability in G-7 economies: Insights from the method of moments quantile regression. *Asia-Pac. J. Reg. Sci.* **2025**, *9*, 743–772. [\[CrossRef\]](#)
33. Machado, J.A.; Silva, J.S. Quantiles via moments. *J. Econom.* **2019**, *213*, 145–173. [\[CrossRef\]](#)
34. Can, M.; Oluc, I.; Sturm, B.; Guzel, I.; Gavurova, B.; Popp, J. Nexus between trading non-green products and environment: Introducing non-green trade openness index. *Front. Environ. Sci.* **2022**, *10*, 950453. [\[CrossRef\]](#)
35. Shahbaz, M.; Nasreen, S.; Ahmed, K.; Hammoudeh, S. Trade openness–carbon emissions nexus: The importance of turning points of trade openness for country panels. *Energy Econ.* **2017**, *61*, 221–232. [\[CrossRef\]](#)
36. Khan, S.M.; Rahman, S.U.; Butt, M. The OIC Sustainability Paradox: How Institutions, Foreign Direct Investment and Innovations Shape Emissions and Ecological Footprints. *J. Political Stab. Arch.* **2025**, *3*, 1173–1208. [\[CrossRef\]](#)
37. Balsalobre-Lorente, D.; Nur, T.; Topaloglu, E.E.; Evcimen, C. The dampening effect of geopolitical risk and economic policy uncertainty in the linkage between economic complexity and environmental degradation in the G-20. *J. Environ. Manag.* **2024**, *351*, 119679. [\[CrossRef\]](#) [\[PubMed\]](#)
38. Ongan, S.; Gocer, I.; Işık, C. Introducing the new ESG-based sustainability uncertainty index (ESGUI). *Sustain. Dev.* **2025**, *33*, 4457–4467. [\[CrossRef\]](#)
39. Alalmaee, H. The Impact of Sustainability Uncertainty on Financial Development. *Sustain. Dev.* **2026**, advance online publication. [\[CrossRef\]](#)
40. Zeren, F.; Dogan, M.; Eryilmaz, S. Determinants of ESG-Related Sustainability Uncertainty: Asymmetric Fourier Evidence from G7 Countries. *Corp. Soc. Responsib. Environ. Manag.* **2026**, *33*, 1479–1492. [\[CrossRef\]](#)
41. DiMaggio, P.J.; Powell, W.W. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *Am. Sociol. Rev.* **1983**, *48*, 147–160. [\[CrossRef\]](#)
42. North, D.C. *Institutions, Institutional Change and Economic Performance*; Cambridge University Press: Cambridge, UK, 1990.
43. Bergougui, B.; Mehibel, S.; Boudjana, R.H. Asymmetric nexus between green technologies, economic policy uncertainty, and environmental sustainability: Evidence from Algeria. *J. Environ. Manag.* **2024**, *360*, 121172. [\[CrossRef\]](#) [\[PubMed\]](#)
44. Alharbey, M.; Ben-Salha, O. Does climate policy uncertainty predict renewable energy stocks? A quantile-based (a) symmetric causality analysis. *Energy Strategy Rev.* **2024**, *54*, 101465. [\[CrossRef\]](#)
45. Kantur, Z.; Ozean, G. Monetary policymaking under climate uncertainty. *Bus. Econ. Res. J.* **2022**, *13*, 579–591. [\[CrossRef\]](#)
46. Kahneman, D.; Tversky, A. Prospect theory: An analysis of decision under risk. In *Handbook of the Fundamentals of Financial Decision Making: Part I*; World Scientific Publishing Co., Ltd.: Singapore, 2013; pp. 99–127. [\[CrossRef\]](#)

47. Ozkan, O.; Adebayo, T.S.; Usman, O. Dynamic connectedness of clean energy markets, green markets, and sustainable markets: The role of climate policy uncertainty. *Energy* **2024**, *303*, 131957. [\[CrossRef\]](#)
48. Pfeffer, J.; Salancik, G. External control of organizations—Resource dependence perspective. In *Organizational Behavior*; Routledge: London, UK, 2015; Volume 2, pp. 355–370.
49. Ren, X.; Li, J.; He, F.; Lucey, B. Impact of climate policy uncertainty on traditional energy and green markets: Evidence from time-varying granger tests. *Renew. Sustain. Energy Rev.* **2023**, *173*, 113058. [\[CrossRef\]](#)
50. Aydoğdu, A.; Dogan, M. Mapping the asymmetric dynamics between ESG uncertainty and clean energy: A quantile–wavelet framework. *J. Environ. Manag.* **2026**, *398*, 128434. [\[CrossRef\]](#) [\[PubMed\]](#)
51. World Bank. Indicators. 2026. Available online: <https://data.worldbank.org/indicator> (accessed on 10 March 2026).
52. Ahir, H.; Bloom, N.; Furceri, D. *The World Uncertainty Index (No. w29763)*; National Bureau of Economic Research: Cambridge, MA, USA, 2022. [\[CrossRef\]](#)
53. Baker, S.R.; Bloom, N.; Davis, S.J. Measuring economic policy uncertainty. *Q. J. Econ.* **2016**, *131*, 1593–1636. [\[CrossRef\]](#)
54. Baltagi, B.H.; Bresson, G.; Pirotte, A. Panel unit root tests and spatial dependence. *J. Appl. Econom.* **2007**, *22*, 339–360. [\[CrossRef\]](#)
55. Pesaran, M.H. *General Diagnostic Tests for Cross Section Dependence in Panels*; Cambridge Working Papers; IZA Discussion Paper No. 1240; CESifo: Munich, Germany, 2004; pp. 1–41. [\[CrossRef\]](#)
56. Pesaran, M.H.; Yamagata, T. Testing slope homogeneity in large panels. *J. Econ.* **2008**, *142*, 50–93. [\[CrossRef\]](#)
57. Pesaran, M.H. A simple panel unit root test in the presence of cross-section dependence. *J. Appl. Econ.* **2007**, *22*, 265–312. [\[CrossRef\]](#)
58. Westerlund, J. Testing for error correction in panel data. *Oxf. Bull. Econ. Stat.* **2007**, *69*, 709–748. [\[CrossRef\]](#)
59. Chudik, A.; Pesaran, M.H. Common correlated effects estimation of heterogeneous dynamic panel data models with weakly exogenous regressors. *J. Econom.* **2015**, *188*, 393–420. [\[CrossRef\]](#)
60. Eberhardt, M.; Bond, S. *Cross-Section Dependence in Nonstationary Panel Models: A Novel Estimator*. MPRA Paper No. 17692; Munich Personal RePEc Archive: Munich, Germany, 2009. Available online: https://mpra.ub.uni-muenchen.de/17692/1/MPRA_paper_17692.pdf (accessed on 10 March 2026).
61. Eberhardt, M.; Teal, F. *Productivity Analysis in Global Manufacturing Production*; Department of Economics, University of Oxford: Oxford, UK, 2010. Available online: <https://ora.ox.ac.uk/objects/uuid:ea831625-9014-40ec-abc5-516ecfbd2118> (accessed on 10 March 2026).
62. Sorrell, S. *The Rebound Effect: An Assessment of the Evidence for Economy-Wide Energy Savings from Improved Energy Efficiency*; Sussex Energy Group, Technology and Policy Assessment Function of the UK Energy Research Centre: Brighton and Hove, UK, 2007.
63. York, R.; McGee, J.A. Understanding the Jevons paradox. *Environ. Sociol.* **2016**, *2*, 77–87. [\[CrossRef\]](#)
64. Aydin, M.; Degirmenci, T.; Yavuz, H. The influence of multifactor productivity, research and development expenditure, renewable energy consumption on ecological footprint in G7 countries: Testing the environmental Kuznets curve hypothesis. *Environ. Model. Assess.* **2023**, *28*, 693–708. [\[CrossRef\]](#)

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